

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: July 2014 - November 2014

Max. Marks: 100**Duration: 3 Hrs****End Semester Exam**

Class: B.E.

Semester: VII

Name of the Course: Soft Computing

Branch: Computer Engineering

Course Code: UCEE704

Instructions:**(1) All Questions are Compulsory.**

Question No.		Max. Marks																																								
Q1(a)	High speed rail monitoring devices sometimes make use of sensitive sensors to measure the deflection of the earth when a rail car passes. These deflections are measured with respect to some distance from the rail car and, hence are actually very small angles measured in microradians. Let a universe of deflection be $A = [1, 2, 3, 4]$ where A is the angle in microradians and let a universe of distances be $D = [1, 2, 3, 4]$ where D is distance in feet. Suppose relation between these two parameters has been determined as follows: <table> <tr> <td></td> <td>D1</td> <td>D2</td> <td>D3</td> <td>D4</td> </tr> <tr> <td>A1</td> <td>1</td> <td>0.3</td> <td>0.1</td> <td>0</td> </tr> <tr> <td>A2</td> <td>0.2</td> <td>1</td> <td>0.3</td> <td>0.1</td> </tr> <tr> <td>A3</td> <td>0</td> <td>0.7</td> <td>1</td> <td>0.2</td> </tr> <tr> <td>A4</td> <td>0</td> <td>0.1</td> <td>0.4</td> <td>1</td> </tr> </table> $R =$ Now let a universe of rail car weights be $W = [1, 2]$, where W is weight in units of 100,000 pounds. Suppose the fuzzy relation of W to A is given by : <table> <tr> <td></td> <td>W1</td> <td>W2</td> </tr> <tr> <td>A1</td> <td>1</td> <td>0.4</td> </tr> <tr> <td>A2</td> <td>0.5</td> <td>1</td> </tr> <tr> <td>A3</td> <td>0.3</td> <td>0.1</td> </tr> <tr> <td>A4</td> <td>0</td> <td>0</td> </tr> </table> $S =$ Using these two relations, find the relation $R^T \circ S = T$ - Using max-min composition - Using max-product composition		D1	D2	D3	D4	A1	1	0.3	0.1	0	A2	0.2	1	0.3	0.1	A3	0	0.7	1	0.2	A4	0	0.1	0.4	1		W1	W2	A1	1	0.4	A2	0.5	1	A3	0.3	0.1	A4	0	0	10
	D1	D2	D3	D4																																						
A1	1	0.3	0.1	0																																						
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Q1(b)	Explain different types of fuzzy inference systems (FIS) in detail.	10																																								
Q2(a)	Design a Fuzzy logic controller for a domestic washing machine with two inputs dirtiness of the load and weight of the laundry and output as amount of detergent used. Use five descriptors for each linguistic variable. Generate a set of rules for control action and diffuzification.	15																																								
Q2(b)	Give the parameterized traingular membership function to model the following as a fuzzy set. “ Tall Person”	5																																								

Q3(a)	Explain Kohonen self-organizing map and RBFN.	10
Q3(b)	Give the basic methods involved in simulated annealing method. Explain how traveling salesperson problem can be solved using simulated annealing method. OR Explain Genetic algorithm with the help of example.	10
Q4(a)	Explain Error Backpropagation training algorithm for multilayer perceptron with supporting diagrams.	14
Q4(b)	Determine the weights after one iteration for Hebbian learning of a single neuron network starting with initial weight vector $W = [1 \ -1 \ 0 \ 0.5]^T$ Use signum bipolar binary activation function. $X1 = [1 \ -2 \ 1.5 \ 0]^T$ $X2 = [1 \ -0.5 \ -2 \ -1.5]^T$ $X3 = [0 \ 1 \ -1 \ 1.5]^T$ and $c = 1$. OR Explain perceptron training algorithm with example.	6
Q5(a)	Explain ANFIS architecture for Sugeno fuzzy model with suitable diagrams.	10
Q5(b)	Explain printed Character recognition using neural network. OR Write short note on inverse kinematics problem.	10